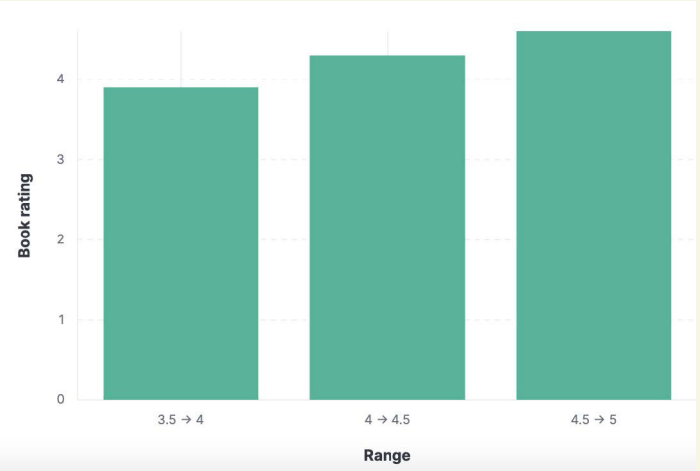
Histogram aggregation with Kibana

In listing 13.10, we developed an aggregation query and executed it in the Kibana Console. The results in JSON format aren't visually appealing, as shown in figure 13.2. It is up to the client who receives that data to represent it as a visual chart. However, Kibana has a rich set of visualizations to aggregate data. Working with the Kibana visualizations is out of scope for this chapter, but this figure shows the same data represented as a histogram in Kibana's Dashboard, this time with an interval of 0.5.



A histogram categorizing books by ratings viewed as a bar chart in Kibana's Dashboard

As you can see, the data is categorized into buckets based on the interval 0.5, and the buckets are filled with the documents that fit in them. To learn about Kibana visualizations, refer to the documentation at https://www.elastic.co/guide/en/kibana/current/dashboard.html.

THE DATE HISTOGRAM

Sometimes we want to group data based on dates rather than numbers. For example, we may want to find all the books released each year, get the weekly sales of an iPhone product, determine the number of server threat attempts every hour, and so forth. This is where the `date_histogram` aggregation comes in handy.

The histogram bucketing strategy we looked at in the last section is based on numerical intervals, but Elasticsearch also provides a histogram based on dates, aptly called `date_histogram`. Let's say we want to categorize books based on their release dates. Here's the query that applies bucketing to do this.

Listing 13.11 A `date_histogram` query

```
GET books/_search
{
  "size":0,
```